

Program 4-30 has two separate variables named `number`. The `cin` and `cout` statements in the inner block (belonging to the `if` statement) can only work with the `number` variable defined in that block. As soon as the program leaves that block, the inner `number` goes out of scope, revealing the outer `number` variable.



WARNING! Although it's perfectly acceptable to define variables inside nested blocks, you should avoid giving them the same names as variables in the outer blocks. It's too easy to confuse one variable with another.

Case Study: See the Sales Commission Case Study on the computer science portal at www.pearsonglobaleditions.com/gaddis.

Review Questions and Exercises

Short Answer

1. Describe the difference between the `if/else if` statement and a series of `if` statements.
2. In an `if/else if` statement, what is the purpose of a trailing `else`?
3. What is a flag and how does it work?
4. Can an `if` statement test expressions other than relational expressions? Explain.
5. Briefly describe how the `&&` operator works.
6. Briefly describe how the `||` operator works.
7. Why are the relational operators called relational?
8. Why do most programmers indent the conditionally executed statements in a decision structure?

Fill-in-the-Blank

9. An expression using the greater-than, less-than, greater-than-or-equal-to, less-than-or-equal-to, equal-to, or not-equal-to operator is called a(n) _____ expression.
10. A relational expression is either _____ or _____.
11. If the values of variables `x` and `y` are 5 and 10, respectively, the relational expression `x<=y` will evaluate to _____.
12. The `if` statement regards an expression with the value 0 as _____.
13. The relational expression `if(-1)` will evaluate to _____.
14. For an `if` statement to conditionally execute a group of statements, the statements must be enclosed in a set of _____.
15. In an `if/else` statement, the `if` part executes its statement or block if the expression is _____, and the `else` part executes its statement or block if the expression is _____.
16. The trailing `else` in an `if/else if` statement has a similar purpose as the _____ section of a `switch` statement.

17. A relational operator has _____ precedence over a binary arithmetic operator.
18. If the subexpression on the left of the _____ logical operator is false, the right subexpression is not checked.
19. If the subexpression on the left of the _____ logical operator is true, the right subexpression is not checked.
20. The _____ logical operator has higher precedence than the other logical operators.
21. A flag variable generally has a data type _____ or _____.
22. The _____ logical operator works best when testing a number to determine if it is within a range.
23. The _____ logical operator evaluates to true if the relational expression is false and evaluates to false if the relational expression is true.
24. A variable with _____ scope is only visible when the program is executing in the block containing the variable's definition.
25. You use the _____ operator to determine whether one `string` object is greater than another `string` object.
26. The value of `a` after execution of the statement `a = 1 ? 0 : 1` will be _____.
27. The _____ in a `switch` block is executed when none of the case values match.
28. The expression following a `case` statement must be a(n) _____.
29. A `break` statement within a block transfers the control to the _____ of the block.
30. What value will be stored in the variable `t` after each of the following statements executes?
 - A) `t = (12 > 1);` _____
 - B) `t = (2 < 0);` _____
 - C) `t = (5 == (3 * 2));` _____
 - D) `t = (5 == 5);` _____

Algorithm Workbench

31. Write an `if` statement that assigns half of `x` to `y` when `x` is even.
32. Write an `if` statement that prints the square root of a number `num` if its value is non-negative, and prints "Error" otherwise.
33. Using the following chart, write an `if/else if` statement that assigns .10, .15, or .20 to `commission`, depending on the value in `sales`.

Sales	Commission Rate
Up to \$10,000	10%
\$10,000 to \$15,000	15%
Over \$15,000	20%

34. Write an `if` statement that sets the variable `hours` to 10 when the flag variable `minimum` is set.
35. Write nested `if` statements that perform the following tests: If `amount1` is greater than 10 and `amount2` is less than 100, display the greater of the two.

36. Write an `if` statement that prints “Age is valid”, if the value of the variable `employee-Age` is within the range 18 through 65, and prints “Invalid Age” otherwise.
37. Write an `if` statement that prints the message “Time is valid” if the variable `hours` is within the range 0 through 23 and the variable `minutes` is within the range 0 through 59.
38. Write a conditional expression that prints “Negative” if the variable `number` is less than 0 and “Positive” otherwise.
39. Assume `str1` and `str2` are `string` objects that have been initialized with different values. Write an `if/else` statement that compares the two objects and displays the one that is alphabetically greatest.
40. Convert the following `if/else if` statement into a `switch` statement:

```

if (choice == 1)
{
    cout << fixed << showpoint << setprecision(2);
}
else if (choice == 2 || choice == 3)
{
    cout << fixed << showpoint << setprecision(4);
}
else if (choice == 4)
{
    cout << fixed << showpoint << setprecision(6);
}
else
{
    cout << fixed << showpoint << setprecision(8);
}

```

41. Match the conditional expression with the `if/else` statement that performs the same operation.

A) `q = x < y ? a + b : x * 2;`

B) `q = x < y ? x * 2 : a + b;`

C) `x < y ? q = 0 : q = 1;`

```

_____ if (x < y)
        q = 0;
    else
        q = 1;

_____ if (x < y)
        q = a + b;
    else
        q = x * 2;

_____ if (x < y)
        q = x * 2;
    else
        q = a + b;

```

True or False

42. T F The `=` operator and the `==` operator perform the same operation when used in a Boolean expression.

43. T F A nested `if/else` block with another `if/else` block facilitates checking of multiple conditions.
44. T F A conditionally executed statement should be indented one level from the `if` statement.
45. T F The compiler reports an error if lines within a block are not indented.
46. T F It's safe to assume that all uninitialized variables automatically start with 0 as their value.
47. T F If an `if` statement is nested within the `if` part of another statement, the block within the inner `if` is executed when both the conditions are false.
48. T F When an `if` statement is nested in the `else` part of another statement, as in an `if/else if`, the only time the inner `if` is executed is when the expression of the outer `if` is true.
49. T F The scope of a variable is limited to the block in which it is defined.
50. T F You can use the relational operators to compare string objects.
51. T F The expression `'a' == 'A' || 0 == 0` is true.
52. T F `y < x` is the same as `x >= y`
53. T F `x >= y` is the same as `(x > y && x = y)`

Assume the variables `x = 5`, `y = 6`, and `z = 8`. Indicate by circling the T or F whether each of the following conditions is True or False:

54. T F `x == 5 || y > 3`
55. T F `x==5 && ( y<=10 || z>10 )`
56. T F `2 != y && z != 4`
57. T F `!(x<5 && z>5)`

Find the Errors

Each of the following programs has errors. Find as many as you can.

```
58. // This program averages 3 test scores.
    // It uses the variable perfectScore as a flag.
    include <iostream>
    using namespace std;

    int main()
    {
        cout << "Enter your 3 test scores and I will ";
            << "average them:";
        int score1, score2, score3,
        cin >> score1 >> score2 >> score3;
        double average;
        average = (score1 + score2 + score3) / 3.0;
        if (average = 100);
            perfectScore = true; // Set the flag variable
        cout << "Your average is " << average << endl;
        bool perfectScore;
        if (perfectScore);
        {
            cout << "Congratulations!\n";
            cout << "That's a perfect score.\n";
            cout << "You deserve a pat on the back!\n";
            return 0;
        }
    }
```

```

59. // This program divides a user-supplied number by another
    // user-supplied number. It checks for division by zero.
    #include <iostream>
    using namespace std;

    int main()
    {
        double num1, num2, quotient;

        cout << "Enter a number: ";
        cin >> num1;
        cout << "Enter another number: ";
        cin >> num2;
        if (num2 == 0)
            cout << "Division by zero is not possible.\n";
            cout << "Please run the program again ";
            cout << "and enter a number besides zero.\n";
        else
            quotient = num1 / num2;
            cout << "The quotient of " << num1 <<
            cout << " divided by " << num2 << " is ";
            cout << quotient << endl;
        return 0;
    }

60. // This program uses an if/else if statement to assign a
    // letter grade (A, B, C, D, or F) to a numeric test score.
    #include <iostream>
    using namespace std;

    int main()
    {
        int testScore;

        cout << "Enter your test score and I will tell you\n";
        cout << "the letter grade you earned: ";
        cin >> testScore;
        if (testScore < 60)
            cout << "Your grade is F.\n";
        else if (testScore < 70)
            cout << "Your grade is D.\n";
        else if (testScore < 80)
            cout << "Your grade is C.\n";
        else if (testScore < 90)
            cout << "Your grade is B.\n";
        else
            cout << "That is not a valid score.\n";
        else if (testScore <= 100)
            cout << "Your grade is A.\n";
        return 0;
    }

```

61. // This program uses a switch-case statement to assign a letter grade (A, B, C, D, or F) to a numeric test score.
- ```
#include <iostream>
using namespace std;

int main()
{
 double testScore;

 cout << "Enter your test score and I will tell you\n";
 cout << "the letter grade you earned: ";
 cin >> testScore;
 switch (testScore)
 {
 case (testScore < 60.0):
 cout << "Your grade is F.\n";
 break;
 case (testScore < 70.0):
 cout << "Your grade is D.\n";
 break;
 case (testScore < 80.0):
 cout << "Your grade is C.\n";
 break;
 case (testScore < 90.0):
 cout << "Your grade is B.\n";
 break;
 case (testScore <= 100.0):
 cout << "Your grade is A.\n";
 break;
 default:
 cout << "That score isn't valid\n";
 }
 return 0;
}
```
62. The following statement should determine if x is not greater than 20. What is wrong with it?
- ```
if (!x > 20)
```
63. The following statement should determine if count is within the range of 0 through 100. What is wrong with it?
- ```
if (count >= 0 || count <= 100)
```
64. The following statement should determine if count is outside the range of 0 through 100. What is wrong with it?
- ```
if (count < 0 && count > 100)
```
65. The following statement assigns (0.1*income) to tax if income is greater than 200,000, otherwise assigns 0 to tax. What is wrong with it?
- ```
tax = (income>200000)?(0.1*income:0);
```

## Programming Challenges

### 1. Absolute of a Number

The absolute of a number is its value without the sign. Write a program that asks the user to enter a number and finds the absolute of the number using the conditional operator.

### 2. Age Check

A company is hiring new recruits who should be within 18 to 28 years of age. Write a program that asks the user to input the current year and an applicant's year of birth. It then displays the age and a message indicating whether the applicant is eligible.

### 3. Magic Dates

The date June 10, 1960 is special because when we write it in the following format, the month times the day equals the year.

6/10/60

Write a program that asks the user to enter a month (in numeric form), a day, and a two-digit year. The program should then determine whether the month times the day is equal to the year. If so, it should display a message saying the date is magic. Otherwise, it should display a message saying the date is not magic.

### 4. Areas of Rectangles

The area of a rectangle is the rectangle's length times its width. Write a program that asks for the length and width of two rectangles. The program should tell the user which rectangle has the greater area, or if the areas are the same.

### 5. Zero Points of a Quadratic Equation

A quadratic equation (from the Latin *quadratus* for "square") is an equation of the form

$$ax^2 + bx + c = 0$$

where  $x$  is the unknown variable, and  $a$ ,  $b$ , and  $c$  are coefficients of the equation. The value of the *discriminant* determines whether a quadratic equation has a real solution or not, and how many solutions it can have. When the discriminant is less than zero, the quadratic equation has no real solution. When the discriminant equals zero, there is exactly one solution; otherwise, there are two solutions. The formula for the discriminant  $D$  is

$$D^2 = b^2 - 4ac$$

Write a program that asks the user to input the coefficients  $a$ ,  $b$ , and  $c$  and then displays the number of real solutions for the corresponding quadratic equation. It is not required to calculate the actual solutions.

### 6. Triangle Construction

In a triangle, the sum of any two sides is always greater than the third side. Write a program that prompts the user to input the length of three lines. It then determines whether the three lines can be used to construct a triangle. If a triangle can be constructed, then compute the area of the triangle.

Note that the area of a triangle can be calculated using the following formula, where

$$s = \frac{(a + b + c)}{2} \text{ and } a, b, \text{ and } c \text{ are the lengths of the sides:}$$

$$\text{area} = \sqrt{s(s-a)(s-b)(s-c)}$$



**VideoNote**  
**Solving the Time  
Calculator  
Problem**

## 7. Time Calculator

Write a program that asks the user to enter a number of seconds.

- There are 60 seconds in a minute. If the number of seconds entered by the user is greater than or equal to 60, the program should display the number of minutes in that many seconds.
- There are 3,600 seconds in an hour. If the number of seconds entered by the user is greater than or equal to 3,600, the program should display the number of hours in that many seconds.
- There are 86,400 seconds in a day. If the number of seconds entered by the user is greater than or equal to 86,400, the program should display the number of days in that many seconds.

## 8. Color Mixer

The colors red, blue, and yellow are known as primary colors because they cannot be made by mixing other colors. When you mix two primary colors, you get a secondary color, as shown here:

When you mix red and blue, you get purple.

When you mix red and yellow, you get orange.

When you mix blue and yellow, you get green.

Write a program that prompts the user to enter the names of two primary colors to mix. If the user enters anything other than “red,” “blue,” or “yellow,” the program should display an error message. Otherwise, the program should display the name of the secondary color that results by mixing two primary colors.

## 9. Middle of Three Numbers

Write a program that lets the user input three numbers and outputs the number whose value is between the other two. This sounds like a very easy program, but it is not as straightforward as it sounds. This is a typical example of a task that is trivial for a human, but one that many people find hard to express in a program. Test at least for the values 1, 2, and 3 and for the values 3, 2, and 1.

## 10. Days in a Month

Write a program that asks the user to enter the month (letting the user enter an integer in the range of 1 through 12) and the year. The program should then display the number of days in that month. Use the following criteria to identify leap years:

1. Determine whether the year is divisible by 100. If it is, then it is a leap year if and only if it is divisible by 400. For example, 2000 is a leap year but 2100 is not.
2. If the year is not divisible by 100, then it is a leap year if and only if it is divisible by 4. For example, 2008 is a leap year but 2009 is not.

Here is a sample run of the program:

Enter a month (1–12): **2**

Enter a year: **2008**

29 days



11. Math Tutor

*This is a modification of Programming Challenge 17 from Chapter 3.* Write a program that can be used as a math tutor for a young student. The program should display two random numbers that are to be added, such as:

$$\begin{array}{r} 247 \\ + 129 \\ \hline \end{array}$$

The program should wait for the student to enter the answer. If the answer is correct, a message of congratulations should be printed. If the answer is incorrect, a message should be printed showing the correct answer.

12. Software Sales

A software company sells a package that retails for \$99. Quantity discounts are given according to the following table.

| Quantity    | Discount |
|-------------|----------|
| 10–19       | 20%      |
| 20–49       | 30%      |
| 50–99       | 40%      |
| 100 or more | 50%      |

Write a program that asks for the number of units sold and computes the total cost of the purchase.

*Input Validation: Make sure the number of units is greater than 0.*

13. Book Club Points

Serendipity Booksellers has a book club that awards points to its customers based on the number of books purchased each month. The points are awarded as follows:

- If a customer purchases 0 books, he or she earns 0 points.
- If a customer purchases 1 book, he or she earns 5 points.
- If a customer purchases 2 books, he or she earns 15 points.
- If a customer purchases 3 books, he or she earns 30 points.
- If a customer purchases 4 or more books, he or she earns 60 points.

Write a program that asks the user to enter the number of books he or she has purchased this month then displays the number of points awarded.

14. Membership Fees

An organization wishing to attract more people decides to base its membership fees on the age of the member. Also, wanting members to attend more activities, it gives a reduction on the membership fee for each activity attended in the previous year.

The following table depicts the corresponding fee and reductions. The minimum membership fee is \$1, even if the member attended a lot of activities.

| Age             | Membership Fee | Reduction per Activity |
|-----------------|----------------|------------------------|
| 6 years or less | \$5            | \$0.75                 |
| 7–12 years      | \$10           | \$1.25                 |
| 13–18 years     | \$15           | \$2                    |
| Over 18 years   | \$25           | \$2                    |

Write a program that asks the user to input their age and the number of activities attended and then displays the corresponding membership fee.

*Input Validation: Do not accept a negative value for either the age or the number of activities.*

### 15. Shipping Charges

The Fast Freight Shipping Company charges the following rates:

| Weight of Package (in Kilograms)   | Rate per 500 Miles Shipped |
|------------------------------------|----------------------------|
| 2 kg or less                       | \$1.10                     |
| Over 2 kg but not more than 6 kg   | \$2.20                     |
| Over 6 kg but not more than 10 kg  | \$3.70                     |
| Over 10 kg but not more than 20 kg | \$4.80                     |

Write a program that asks for the weight of the package and the distance it is to be shipped, then displays the charges.

*Input Validation: Do not accept values of 0 or less for the weight of the package. Do not accept weights of more than 20 kg (this is the maximum weight the company will ship). Do not accept distances of less than 10 miles or more than 3,000 miles. These are the company's minimum and maximum shipping distances.*

### 16. Negative Split

Running the second half of a race faster than the first half is called a negative split. Write a program that asks for the time (in seconds) for the first half of a race and for the total of the race, and outputs a congratulatory message in case of a negative split.

*Input Validation: Do not accept negative values for either time. Also check that the second time is greater than the first time.*

### 17. Toss of a Coin

Write a program that simulates the toss of a coin. The program asks the user to input either H (for head) or T (for tail). It then randomly generates either H (internally represented by 1) or T (internally represented by 0). If the user's prediction matches with the generated toss, display "Won"; otherwise display "Lost".

### 18. Fat Gram Calculator

Write a program that asks for the number of calories and fat grams in a food. The program should display the percentage of calories that come from fat. If the calories from fat are less than 30 percent of the total calories of the food, it should also display a message indicating that the food is low in fat.

One gram of fat has 9 calories, so

Calories from fat = fat grams \* 9

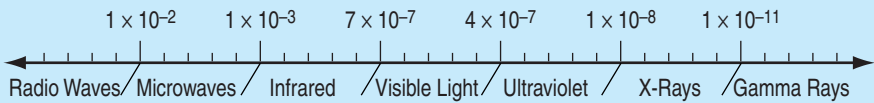
The percentage of calories from fat can be calculated as

Calories from fat ÷ total calories

*Input Validation: Make sure the number of calories and fat grams are not less than 0. Also, the number of calories from fat cannot be greater than the total number of calories. If that happens, display an error message indicating that either the calories or fat grams were incorrectly entered.*

19. Spectral Analysis

If a scientist knows the wavelength of an electromagnetic wave, he or she can determine what type of radiation it is. Write a program that asks for the wavelength of an electromagnetic wave in meters and then displays what that wave is according to the chart below. (For example, a wave with a wavelength of 1E-10 meters would be an X-ray.)



20. The Speed of Sound

The following table shows the approximate speed of sound in air, water, and steel.

| Medium | Speed                  |
|--------|------------------------|
| Air    | 1,100 feet per second  |
| Water  | 4,900 feet per second  |
| Steel  | 16,400 feet per second |

Write a program that displays a menu allowing the user to select air, water, or steel. After the user has made a selection, he or she should be asked to enter the distance a sound wave will travel in the selected medium. The program will then display the amount of time it will take. (Round the answer to four decimal places.)

*Input Validation:* Check that the user has selected one of the available choices from the menu. Do not accept distances less than 0.

21. The Speed of Sound in Gases

When sound travels through a gas, its speed depends primarily on the density of the medium. The less dense the medium, the faster the speed will be. The following table shows the approximate speed of sound at 0 degrees centigrade, measured in meters per second, when traveling through carbon dioxide, air, helium, and hydrogen.

| Medium         | Speed (Meters per Second) |
|----------------|---------------------------|
| Carbon dioxide | 258.0                     |
| Air            | 331.5                     |
| Helium         | 972.0                     |
| Hydrogen       | 1,270.0                   |

Write a program that displays a menu allowing the user to select one of these four gases. After a selection has been made, the user should enter the number of seconds it took for the sound to travel in this medium from its source to the location at which it was detected. The program should then report how far away (in meters) the source of the sound was from the detection location.

*Input Validation:* Check that the user has selected one of the available menu choices. Do not accept times less than 0 seconds or more than 30 seconds.

## 22. Freezing and Boiling Points

The following table lists the freezing and boiling points of several substances. Write a program that asks the user to enter a temperature then shows all the substances that will freeze at that temperature, and all that will boil at that temperature. For example, if the user enters  $-20$ , the program should report that water will freeze and oxygen will boil at that temperature.

| Substance     | Freezing Point ( $^{\circ}\text{F}$ ) | Boiling Point ( $^{\circ}\text{F}$ ) |
|---------------|---------------------------------------|--------------------------------------|
| Ethyl alcohol | $-173$                                | 172                                  |
| Mercury       | $-38$                                 | 676                                  |
| Oxygen        | $-362$                                | $-306$                               |
| Water         | 32                                    | 212                                  |

## 23. Guessing Games

Write a program that displays the following menu:

Guessing games

1. Play Higher or Lower
2. Play paper-scissors-rock
3. Guess the number of fingers
4. Quit

Enter your choice (1–4):

If the user enters 1, the program should generate two random numbers between 1 and 20 and display the first number. It should then ask the user to enter either H or L for Higher or Lower. The user wins if they entered H and the second number was higher than the first or when they entered L and the second number was lower.

If the user enters 2, the program should generate one random number between 1 and 3 with 1 corresponding to paper, 2 to scissors, and 3 to rock. The program should then ask the user to enter P for paper, S for scissors, or R for rock. The program should then display who won: the computer or the user. Scissors beats paper, rock beats scissors, and paper beats rock.

Finally, if the user enters 3, the program should generate a random number between 1 and 5. The user can enter a number in the same range, and they win when their guess is correct.

*Input Validation: Display an error message if the user enters a number outside the range of 1 through 4 when selecting an item from the menu. Do not accept characters other than H or L for Higher/Lower, other than P, S, and R for paper-scissors-rock, or a value outside the range from 1 to 5 for guessing fingers.*

## 24. Parking Lot

A parking lot charges the following rates for parking:

| Number of Hours            | Rate per Hour |
|----------------------------|---------------|
| $\leq 2$                   | \$2           |
| Over 2 but not more than 4 | \$1.75        |
| $\geq 5$                   | \$1.50        |

Write a program that asks for the starting and ending times of a parked car and displays the amount to be paid. The program should ask for the times to be entered as floating-point numbers in the form HH.MM. For example, 07:00 hours will be entered as 07.00, and 16:28 hours will be entered as 16.28.

Note that each initiated hour must be paid in full. For example, 1 hour and 3 minutes will cost \$4.

*Input Validation: The program should not accept times that are greater than 21:59 or less than 07:00, the operating hours of the parking lot. Also, no number whose last two digits are greater than 59 should be accepted.*

*Hint: You can get the minutes of the time by multiplying the time by 100 and then taking the remainder of the division by 100.*

## 25. Tourist Attraction

A tourist attraction offers three types of tickets:

- Regular adult tickets at \$30 per person, with a reduction of 10% for groups of at least five persons.
- Student tickets at \$25 per person
- Family tickets for two adults and two or more children. The price is \$75, with an additional \$15 for each child beyond two children.

Write a program that calculates the total price of tickets. The program displays a menu with the three options, lets the user select an option, and then asks for the number of people in the party. It should then display the total amount due.

*Input Validation: Make sure the user cannot select invalid options in the menu and that the number of people in the party is always greater than 0.*

## 26. Tourist Attraction, Part 2

Extend the previous solution by showing at the end of the program how much money was saved compared to a regular adult ticket. If there are no savings, no message should be shown. Also print out an advertisement for a “Come back” ticket at \$10 each and include the total amount in the message.

## 27. Wi-Fi Diagnostic Tree

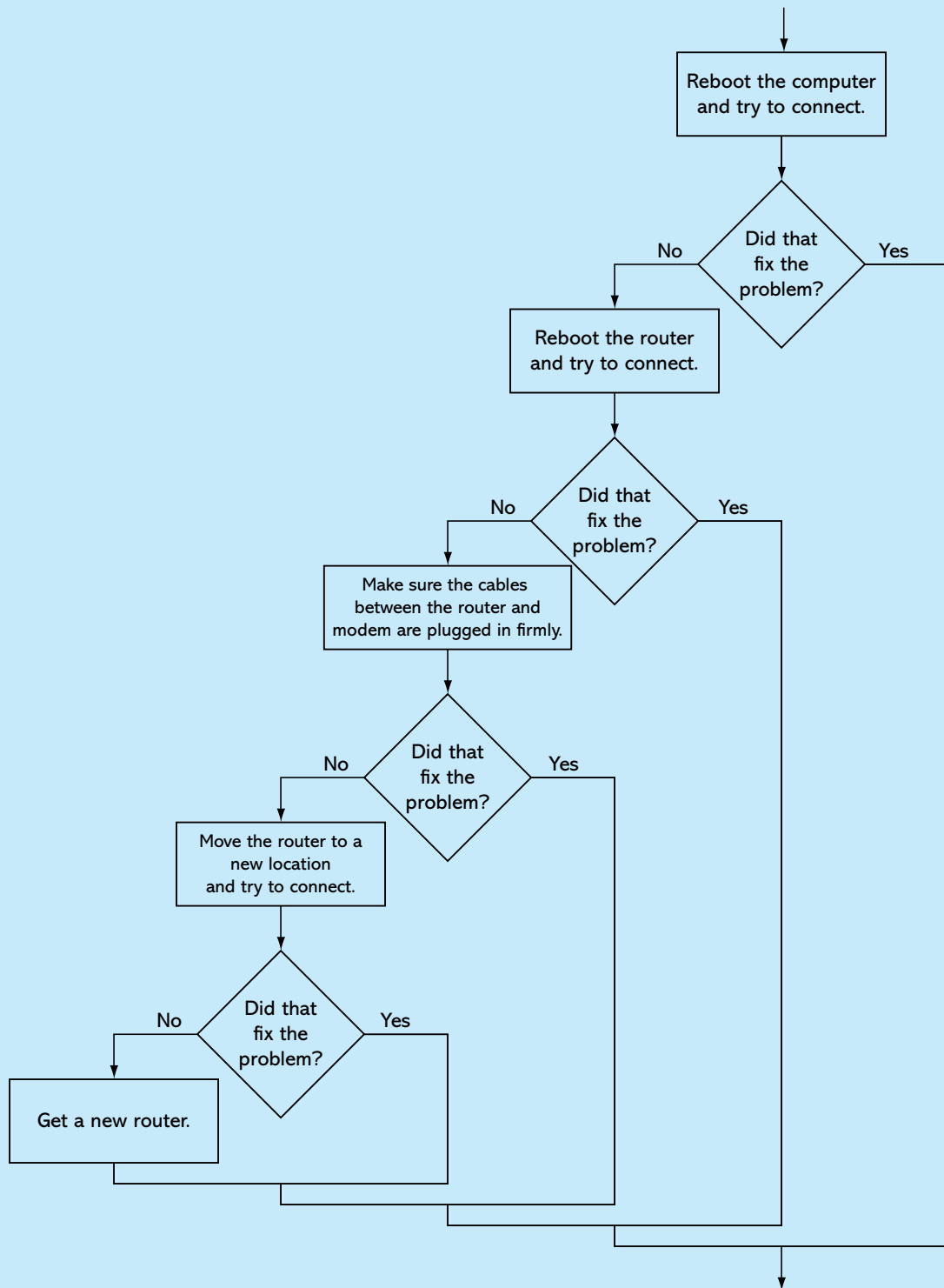
Figure 4-11 shows a simplified flowchart for troubleshooting a bad Wi-Fi connection. Use the flowchart to create a program that leads a person through the steps of fixing a bad Wi-Fi connection. Here is an example of the program’s output:

```
Reboot the computer and try to connect.
Did that fix the problem? no
Reboot the router and try to connect.
Did that fix the problem? yes
```

Notice the program ends as soon as a solution is found to the problem. Here is another example of the program’s output:

```
Reboot the computer and try to connect.
Did that fix the problem? no
Reboot the router and try to connect.
Did that fix the problem? no
Make sure the cables between the router and modem are plugged in firmly.
Did that fix the problem? no
Move the router to a new location.
Did that fix the problem? no
Get a new router.
```

**Figure 4-11** Troubleshooting a bad Wi-Fi connection



## 28. Exercise Generator

Write a program that generates a math exercise for the user.

The program should first ask for which operation an exercise should be generated. The possibilities are addition (+), subtraction (−), multiplication (\*), and division (/).

After selecting an operation, the user should answer the following question:

What is the maximum value for the input values of the exercise?

The answer to this question must be stored in a variable `maximumValue`.

Next, the program should ask

Are negative values allowed in the exercise?

The possible answers for the question are “Y” and “N”.

The program will then generate two random numbers. If the user allows negative values, the random numbers must be generated in the range from `-maximumValue` to `+maximumValue`; if the user allows only positive values, the range is restricted from 0 to `maximumValue`. Moreover, if only positive numbers are allowed and subtraction was chosen, the second random value must be less than the first random value.

Finally, the program will show the generated exercise, depending on the chosen operation and the numbers randomly generated, and ask the user for the correct solution. If the user solves the exercise correctly, the program will print out a congratulatory message.